

STUDENT PERFORMANCE PREDICTION SYSTEM

AI/ML Internship Project Report

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I sincerely thank CODEVEDX for providing me with the opportunity to complete this internship project. It enhanced my understanding of Python programming, data analysis and Machine Learning.

Abstract

This project predicts students' academic performance using Machine Learning techniques. It demonstrates data loading, preprocessing, visualization, model training, evaluation and prediction in a console-based application.

Objectives

- Develop a prediction system.
- Analyze student data.
- Visualize academic performance.
- Train and evaluate a Machine Learning model.
- Strengthen Python programming skills.

Problem Statement

Educational institutions require an efficient way to estimate student performance so that timely academic guidance can be provided.

Existing System

Traditional methods rely on manual analysis, making performance prediction slow and less efficient.

Proposed System

The proposed system uses a Machine Learning model to analyze academic factors and predict final examination performance.

Software Requirements

Operating System: Windows 10/11

Language: Python 3.x

IDE: Visual Studio Code

Libraries: Pandas, Matplotlib, Scikit-learn

Technologies Used

- Python
- Pandas
- Matplotlib
- Scikit-learn
- CSV Files

Project Workflow

Load Dataset

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Analyze Data

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Visualize Data

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Train Model

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Predict Performance

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Display Results

Features

- Menu-driven console
- Dataset information
- Visualization
- Model training
- Performance prediction
- Exception handling

Conclusion

The project successfully demonstrates the application of Machine Learning for predicting student performance while improving practical knowledge of Python and data analysis.

References

- Python Documentation
- Pandas Documentation
- Matplotlib Documentation
- Scikit-learn Documentation